

FINAL EXAMINATION

Students must complete this assignment in groups as registered.

Recommendation

This is an open-book assignment, and thus it may include some topics beyond what is provided in lectures. Try your best to find your solution from the vast information on the Internet.

Examinees should use visualization (e.g., figures, tables, etc.) to demonstrate their ideas more intuitively.

Regulations

Provide references for everything that is not from your own

Do not copy or translate the original content. Do not copy other teams' works.

That is plagiarism, which gives you a zero course credits and more troubles.

Grading rubric

Report: 20 points

Part A (10 credits)					Part B (10 credits)		
1	2	3	4	5	1	2	3
2	2	2	2	2	4	3	3

Interview: 10 points

1. Fundamental knowledge

1.1. (2pts) Local Binary Patterns (LBP)

- a. Present the original local binary pattern $LBP_{P,R}$ (0.5pt)
 - (i) State its key idea and operator formula.
 - (ii) Calculate the $LBP_{P,R}$ values on an example image patch.
- b. Which limitation of $LBP_{P,R}$ is addressed by $LBP_{P,R}^{riu}$? Which limitation of $LBP_{P,R}$ is addressed by $LBP_{P,R}^{riu2}$? (0.5pt)
- c. Find a variant of LBP proposed in some reliable publication *after 2015* and present the following points (1pt)
 - (i) State its crucial idea, operator formula, and field of application.
 - (ii) What are this LBP variant's advantages over the original LBP?
 - (iii) Cite the corresponding part in the paper to show that this LBP variant is theoretically or empirically equivalent/superior to some other approaches.

1.2. (2.5pts) Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA)

- a. Demonstrate the steps of Principal Component Analysis via a numerical example (0.5pt)
- b. Demonstrate the steps of Linear Discriminant Analysis via a numerical example (0.5pt)
- c. Pick one dimensionality reduction method in the following [link](#) and present your understanding of the selected method. (1pt)
- d. Compare PCA/LDA with the method you studied in c. (0.5pt)

1.3. (2.5pts) Support Vector Machine (SVM)

- a. Present the steps of the Support Vector Machine to perform linear classification (0.5pt)
- b. What is the use of non-linear kernels? List at least three such kernels (0.5pt)
- c. The original SVM is only for binary classification. However, there are ways to extend SVM to multi-class classification. Present an approach for multi-class classification without reducing the single multi-class problem into multiple binary classification ones. (1pt)

1.4. (3pts) Artificial neural networks

- a. What does the Universal Approximation Theorem state about artificial neural networks? (0.5pt)
- b. What is the vanishing gradient problem? Describe two ways to alleviate the issue (0.5pt)
- c. Briefly describe two neural network architectures. The selections should not be too trivial, and they must not be what is introduced in the lectures (e.g., traditional MLP, vanilla CNN, and vanilla LSTM, etc.) (1pt)
- d. What are optimizers in deep networks? Describe two common optimizers for deep networks. (1pt)

2. Advanced knowledge

2.1. (4pts) Attention mechanisms and transformers

- a. What is attention in machine learning? Then, what does “self-attention” mean? (1pt)
- b. What is a transformer? Describe the components of a transformer. (1pt)
- c. Find a reliable publication **after 2020 in computer science** that uses the attention (or its variants) and present the following points (1pt)
 - (i) Describe how the proposed method implements the attention mechanism.
 - (ii) Cite the corresponding part in the paper to show that the use of attention helps boost the proposed method's performance over other approaches.
- d. Find **another** reliable publication **after 2020 in computer science** that uses the transformer (or its variants) and present the following points (1pt)
 - (i) Describe how transformer is implemented in the proposed method.
 - (ii) Cite the corresponding part in the paper to show that the transformer helps boost the proposed method's performance over other approaches.

2.2. (3pts) Capsule neural networks

- a. What is a capsule neural network? Describe the component of a capsule network (1pt).
- b. In some aspects, capsule neural networks are better than conventional convolutional neural networks. What are these aspects? (0.5pt)
- c. However, the popularity of capsule neural networks has not been grown as expected. What hinders them? (0.5pt)
- d. Find a reliable publication **after 2020 in computer science** that uses the capsule neural network and present the following points (1pt)
 - (i) Describe how the proposed method implements the capsule neural network.
 - (ii) Cite the corresponding part in the paper to show that the use of attention helps boost the proposed method's performance over other approaches.

2.3. (3pts) ChatGPT

- a. Summarize the history of the development of GPT models (0.5pt).
- b. List the strong points and weak points of ChatGPT, and for each point, give an example and provide a snapshot of the ChatGPT inference as evidence. (1pt)
- c. Identify the algorithms and techniques implemented in ChatGPT. For each strategy, clearly explain its effects to the performance of ChatGPT. (1pt)
- d. Which issues are primarily concerned about ChatGPT? (0.5pt)